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Enhancing construction safety: Machine learning-based classification of injury types

Maryam Alkaissy [a], Mehrdad Arashpour [a,*], Emadaldin Mohammadi Golafshani [a], M. Reza Hosseini [b], Sadegh Khanmohammadi [a], Yu Bai [a], Haibo Feng [c]

[a] Department of Civil Engineering, Monash University, Melbourne, Australia
[b] School of Architecture & Built Environment, Deakin University, Victoria, Australia
[c] Faculty of Forestry, University of British Columbia, Canada

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ABSTRACT

The construction industry is a hazardous industry with significant injuries and fatalities. Few studies have used data-driven analysis to investigate injuries due to construction accidents. This study aims to deploy machine learning (ML) models to predict four injury types (ITs): Upper limbs, lower limbs, head/neck, and back/trunk. A total of 16,878 construction accident records in Australia were collected and fed into several ML algorithms, including fine trees, ensemble of boosted trees, xgboost, random forest, two types of support vector machines, and logistic regression. Six performance metrics of precision, recall, accuracy, F1 score, the area under the receiver operating curve (AUROC), and the area under precision recall curve (AUPRC) were used to evaluate modeling outputs. Random forest showed superior performance in predicting injury types (accuracy 79.3%; recall 78.0%; F1 score 78.5%; precision 77.1%; AUROC 0.98; and AUPRC 0.78). The critical features of injury types were analyzed using the feature importance method and accident nature and mechanism had significant impacts. The study's findings contribute to safety enhancement by providing quantitative prediction models of injury types and subsequent development of safety controls in construction.

1. Introduction

Construction sites are hazardous environments, and accidents frequently cause serious injuries (Xu et al., 2019; Priemus and Ale, 2010). These injuries impact workers physically and emotionally, especially when complicated by chronic medical conditions (Xu and Zou, 2021; Alkaissy et al., 2020). In many cases, injured workers are incapacitated, and construction companies are exposed to consequences in terms of compensation costs and negative publicity (Alkaissy et al., 2021). Therefore, understanding the contributors to different injury types is crucial for an informed and proactive safety management plan (Arashpour et al., 2022).

In 2020, Safe Work Australia released its latest Occupational Accidents Statistical Report (Australia, 2021), with the average distribution of injury types (IT) in the construction industry over the past years (see Fig. 1). This report includes the distribution of injury types witnessed in different industry sub-sectors, demographics, and temporal distributions considering several accident mechanisms. As illustrated in Fig. 1, the most frequent type of injury in construction worksite accidents was to the upper limbs of injured workers, accounting for 25 % of injuries. These injuries affected injured workers' welfare and mobility, and resulted in high compensation costs and scheduled downtimes for construction projects (Xu and Zou, 2021). In addition, injuries to workers' heads and necks were more likely to be fatal than other injury types (Dumrak et al., 2013). While 23 % of injuries occurred to lower limbs, the second most frequent injury type, these injuries were less likely to have severe consequences than other injury types (Passmore et al., 2019).

Few studies have investigated the factors contributing to injuries to different parts of workers' bodies due to construction accidents. The lack of in-depth understanding and the shortage of data hinders the development of models to predict injury types to enable organizations to identify areas for safety improvements (Arashpour et al., 2021; Alkaissy et al., 2022). The main reasons for this shortage are the lack of a

* Corresponding author at: Monash University, Department of Civil Engineering, Vic 3800, Australia.
E-mail addresses: maryam.alkaissy@monash.edu (M. Alkaissy), mehrdad.arashpour@monash.edu (M. Arashpour), emad.golafshani@monash.edu (E.M. Golafshani), reza.hosseini@deakin.edu.au (M.R. Hosseini), sadegh.khanmohammad@monash.edu (S. Khanmohammadi), yu.bai@monash.edu (Y. Bai), haibo.feng@ubc.ca (H. Feng).

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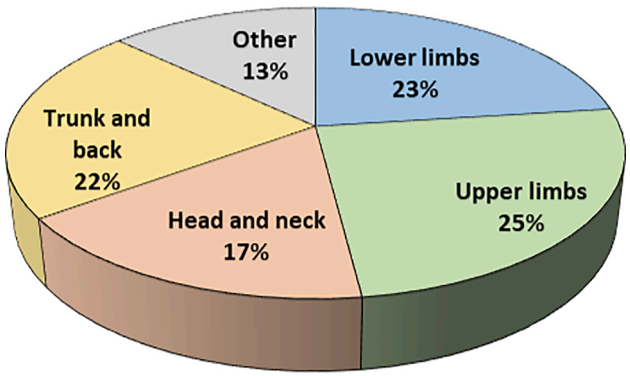

Fig. 1. Construction workers' injury types due to worksite accidents in Australia (2016–2019).

sufficiently large injury data set and/or the poor data quality, making it difficult to extract useful information to address the complex consequences of construction injuries (Elfahham, 2019). Moreover, most studies have approached this problem using statistical analysis tools through which it is usually challenging to interpolate complex information and patterns of injury-type data (Kattan and Gerds, 2020; Al-Kaissy et al., 2019).

Providing a quantitative predictive model and understanding the contributing features that determine injury types in the construction industry will aid decision-makers in enhancing construction safety and developing efficient controls (Ramaswamy and Mosher, 2018). A predictive ML model can provide insights to develop safety enhancement strategies to improve the construction industry's overall safety performance (Tang and Golparvar-Fard, 2021; Alkaissy et al., 2021). Thus, this study aims to use ML to develop an entirely data-driven predictive model of injury type due to construction worksite accidents by mining an extensive construction injury data set. ML models allow organizations to accurately estimate the likely injury types due to construction worksite accidents. However, choosing optimum ML algorithms to predict injury types in construction settings is nontrivial. Therefore, the first research question in this study is conceived:

Which ML algorithm(s) can accurately predict injury types due to construction worksite accidents?

An in-depth understanding of the features that determine injury types, such as the age, gender, and the nature/mechanism of the injury, is essential in enhancing construction safety. Features contributing to injury types are pivotal to developing efficient safety controls and protective measures (Xu and Zou, 2021; Ramaswamy and Mosher, 2018; Berglund et al., 2019). Consequently, the second research question in this study is formulated:

What are the most significant accident features in predicting injury types using ML algorithms?

This paper is structured as follows: Section 2 presents a brief review of relevant literature. A detailed explanation of the research methods, including data description and data pre-processing, is presented in Section 3. Section 4 presents the ML methods results. Section 5 discusses the findings obtained using ML models. The final section (Section 6) presents the study's conclusions, highlighting its contributions and outlining future research directions.

2. Background

Analysis of the features contributing to construction accidents has been receiving greater attention in the past decade. Zhang and Hassan (Zhang and Hassan, 2019) investigated the significant factors contributing to work zone accidents in the Egyptian construction industry from 2010 to 2016, concluding that age and gender were the main factors, besides some work-related characteristics. Sharwood, et al. (Sharwood et al., 2018) studied the attributes of work-related traumatic spinal injuries in Australia and found that the mean age of injured workers was 46.6 years. Of these injuries, 50 % are related to falls from heights, with the frequency of injuries in the construction industry. Chen, et al. (Chen et al., 2017) investigated the role of demographics, the size of the employer and accident mechanisms in the Canadian construction industry with both organizational and individual factors highlighted as important aspects in determining the individual's safety performance and psychological comfort.

Traditional statistical analysis models have been widely used to analyze site accidents in construction and other complex industries (Arashpour et al., 2019; Alkaissy et al., 2021). However, ML techniques have a high potential for predicting future accident events (Kwon et al., 2021). Machine learning (ML) models, developed for events classification and prediction, have been widely implemented in areas such as finance (Dixon et al., 2020; Renault, 2020; Ghoddusi et al., 2019) and education (Arashpour et al., 2019; Arashpour et al., 2020; Arashpour et al., 2022), as well as construction (Tixier et al., 2016; Nath et al., 2018; Bilal and Oyedele, 2020).

The powerful ML capabilities reveal patterns and relationships between the different contributing features in a system, leading to an informed decision-making process (Xu and Saleh, 2021). One of the challenges that ML methods can address and on which they can provide insightful analysis is safety in construction. Machine learning (ML) methods have recently been used to overcome traditional statistical models' limitations in construction safety analyses, such as specific assumptions on data and model configurations (Kattan and Gerds, 2020). Historical data of accidents in construction can serve as an important feed into ML techniques to predict injury types of construction worksite accidents and to provide logical links to contributing features.

Machine learning (ML) techniques have recently received much attention, especially in investigating and predicting accidents in different sectors, such as transportation, manufacturing, and construction (Chen et al., 2016). However, ML in construction accident analysis continues to be much less researched compared to other sectors. Using a support vector machine (SVM) and an artificial neural network (ANN), Sarkar, et al. (Sarkar et al., 2019) predicted the severity of construction accidents on injured workers using two attributes, that is, accident mechanisms and day and month of the injury, both of which were deemed important for achieving an accurate accident prediction model. Their study's findings revealed that the pre-processing task, known as the SVM classifier parameter optimization, was the key to constructing an efficient prediction model. Kang and Ryu (Kang and Ryu, 2019) used the random forest (RF) method to predict and investigate important factors determining accident types in construction. Their findings revealed that accidents were primarily triggered by human errors, inadequate supervision, and a shortage of safety facilities and personal protective equipment (PPE). Ramaswamy and Mosher (Ramaswamy and Mosher, 2018) used workers' compensation claims to measure injury severity, investigating the effects of 'age', 'tenure', 'type of claim', 'body part injured', and 'nature and cause of injury' on work-related injury costs. Their results indicated that factors such as 'age', 'tenure of employee', 'type of claim', 'body part injured', 'nature of injury' and 'cause of injury' have a substantial impact when determining compensation claim amounts. Cheng, et al. (Cheng et al., 2012) analyzed the causes of injuries and fatalities due to falls and collapses using a classification and regression tree (CART), finding that factors linked and directly related to worksite accidents comprised 'source of injury', 'project type', 'accident type' and 'enterprise size'. Table 1 presents a summary of relevant studies in the literature and contributing features of work-related accidents, with this table highlighting the most common attributes of construction worksite accidents.

Machine learning (ML) methods comprise a set of different algorithms from which practitioners can choose based on the type of problem and the data to be analyzed (Xu and Saleh, 2021). The support vector machine (SVM) is an ML algorithm widely used in construction

Table 1
Summary of reviewed literature.

Paper	Employer size	Sub-sector	Agency group	Age	Gender	Hours worked/ week	Employment type	Month	Day	Accident mechanism	Accident nature	Method/s
(Koc et al., 2021)		X								X	X	DT
(Sarkar et al., 2020)								X	X	X	X	SVM/ANN
(Choi et al., 2020)	X	X		X	X				X			DT, LR, RF
(Birhane et al., 2020)				X								Cause and effect
(Zhang and Hassan, 2019)		X	X	X	X			X	X			Multivariate analysis
(Kang and Ryu, 2019)	X		X	X			X			X	X	RF
(Winge et al., 2019)		X		X						X		Cause and effect risk assessment
(Ramdan et al., 2019)				X		X	X			X		Cramer's V test
(Wong et al., 2019)		X	X						X			LR
(Ramaswamy and Mosher, 2018)			X	X			X		X	X	X	Chi-sq
(Khashaba et al., 2018)			X	X						X	X	Case control study
(Guo et al., 2018)	X			X	X							Multi group factor analysis
(Winge and Albrechtsen, 2018)		X		X	X					X		Descriptive epidemiology
(Sharwood et al., 2018)		X	X	X	X			X		X	X	Descriptive statistics
(Chen et al., 2017)	X			X	X	X				X		CART
(Khodabandeh et al., 2016)				X	X		X			X	X	Pearson test & Kruskal-Wallis
(Soltanzadeh et al., 2016)	X	X		X								Chi-sq
(Hosseini et al., 2015)	X			X				X				Data visualization report
(Alizadeh et al., 2015)				X								BN
(Suárez-Cebador et al., 2014)		X	X	X						X	X	Chi-sq
(Lipscomb et al., 2014)		X		X								Descriptive statistics
(Dumrak et al., 2013)	X	X		X	X			X	X	X	X	Descriptive statistics
(Cheng et al., 2012)	X	X		X	X		X			X		CART
(Cheng et al., 2010)	X	X	X	X	X					X		Characteristic analysis
(Barss et al., 2009)			X	X						X		Injury severity score analysis

Notes: **CART** = Classification and regression tree, **RF** = Random Forest, **LR** = Logistic regression, **SVN** = Support vector machine, **ANN** = Artificial neural network, **BN** = Bayesian network, **DT** = Decision tree.

safety analysis predictions due to its sound projecting performance and superior theoretical foundation. In addition, the decision tree (DT) is a smooth ML method used to discover hidden patterns due to the independence of attributes assumption, with the potential to deal with qualitative and quantitative attributes in a data set (Sarkar et al., 2019). This makes the decision tree method a widely used ML approach in different fields, including medicine (Madadipouya, 2015), construction (Mistikoglu et al., 2015), and management (Dey, 2012).

Although studies have been conducted on construction accidents using different analytical tools, few studies have used ML methods to predict injury types caused by construction worksite accidents, a gap this study aims to address. Predicting injury types is vital for providing adequate personal protective equipment (PPE) at the individual level, and deliberate safety management practice at the organizational level. The study's findings can contribute to an in-depth understanding of features predicting injury types using ML algorithms and provide insights for enhancing construction safety using data-driven modeling.

3. Materials and method

This study takes four key steps to develop predictive models: data handling and description, pre-processing data to feed ML algorithms, running algorithms and tuning parameters, and applying performance and evaluation metrics to the prediction models. The research steps and processes are elaborated in Fig. 2 and the following sub-sections.

3.1. Data handling and description

The data set of construction accidents in Victoria, Australia, between 2016 and 2021 was obtained from Safe Work Australia, a governmental

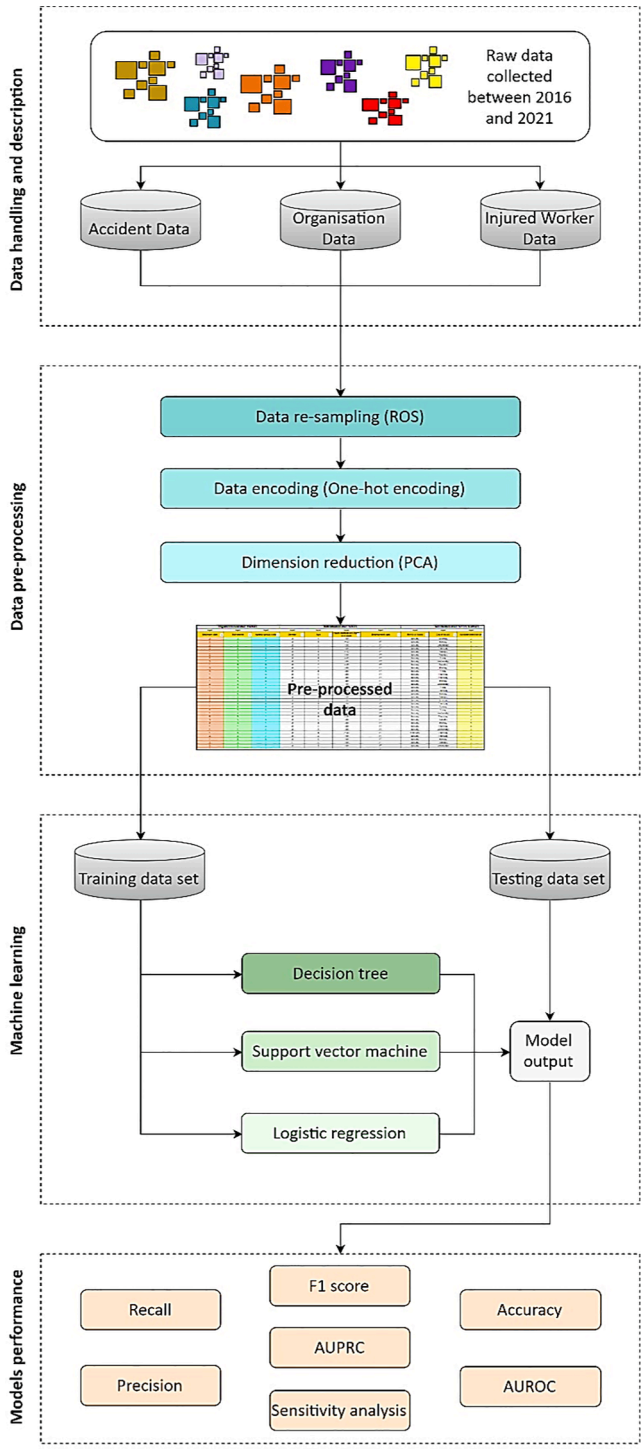

Fig. 2. Research workflow.

statutory agency that collects and reports occupational accident data and develops and evaluates national policy strategies and the legislative framework. Attributes in the obtained data set were thoroughly investigated to highlight the features directly prominent in predicting construction injury types resulting from worksite accidents. As a result, 11 contributing features were revealed to predict the outcome of worksite accidents on four injury types (see Table 2).

Table 2 summarizes the input and output attributes used in this research. The predictor attributes were categorized into three groups reflecting the level of analysis: Group 1 comprised 'employer size', 'project type' and 'construction sub-sector', that is, the contributing features related to the organizational level of analysis. Group 2 was related to the individual level of analysis with 'age', 'gender', 'employment type', and 'hours worked pre-injury'. Group 3 comprised other accident-related attributes, such as the 'month of injury', 'day of injury', 'accident mechanism', and 'accident nature'. The data set also consisted of two variable types: categorical (i.e., employer size, sub-sector, project type, age, month, day, accident mechanism, accident nature, and injury type), and binary (i.e., gender, total hours worked pre-injury and employment type). After careful consideration, the 11 features used to predict injured workers' injury types were chosen based on a review of published papers on accident analysis in construction worksites. For example, Sarkar et al. (Sarkar et al., 2020) investigated the effect of 'month' and 'day' of an accident and proved the significance of worksite injuries' severity. Dumrak, et al. (Dumrak et al., 2013) highlighted the importance of 'organization type', 'project type', and 'sub-sector' as critical variables of the accident model. Khodabandeh, et al. (Khodabandeh et al., 2016) also indicated a significant relationship between socio-demographic features and work-related variables in accident prediction models.

Table 2
Contributing features to injuries caused by construction accidents.

Level of analysis	Factors	Attributes	Type code
Organization	Employer size	Small, Medium, Large	Categorical
	Sub-sector	Residential, Non-residential, Heavy civil, Land preparation & site development, Structural services, Installation services, Completion services	Categorical
	Agency type	Machinery, Mobile plant, Powered equipment, Unpowered tools, Materials & substances, Environmental, Animal Agencies	Categorical
Individual	Age	15–19, 20–29, 30–39, 40–49, 50–59, 60+	Categorical
	Gender	Female, Male	Binary
	Total hours worked per week, pre-injury	Less than 40 h, more than 40 h	Binary
	Employment type	Full-time, Part-time	Binary
Other accident-related	Month of injury	Jan, Feb, March, April, May, Jun, Jul, Aug, Sep, Oct, Nov, Dec	Categorical
	Day of injury	Mon, Tues, Wed, Thurs, Fri, Sat, Sun	Categorical
	Accident mechanism	Falls, Hitting objects, Being hit by moving object, Sound & pressure, Body stressing	Categorical
	Accident nature	Fracture, Wounds & lacerations, Muscle/joint/ ligament injury, Musculoskeletal system, Digestive and Circulatory system, Burns	Categorical
Output variable	Injury type	Lower limbs, Upper Limbs, Head & neck, Trunk & back	Categorical

3.2. Data pre-processing

Almost all data-driven projects require significant data pre-processing to obtain high-quality and reliable prediction results. As a key phase of this research, data pre-processing comprised three main steps, as shown in Fig. 2 and discussed in the following sections. These steps comprised data re-sampling, data encoding, and dimension reduction.

3.2.1. Data cleaning and re-sampling

In the first step of data pre-processing, the data set underwent extensive data cleaning, which involved removing missing data, duplications, inconsistencies and outlier data. As a result, a final total of 16,878 construction worksite accidents was achieved, of which 'upper limbs' accidents comprised 41.5 % of the total number of accidents, 'lower limbs' comprised 26.5 %, 'head and neck' injuries were 9.4 %, and accidents causing 'trunk and back' injuries were 22.6 %. The four output classes were particularly imbalanced (i.e., 9.4 % 'head and neck' injuries compared to 41.5 % 'upper limbs' injuries). Thus, data re-sampling was a crucial step to ensure a balanced class distribution (Choi et al., 2020).

3.2.2. Data encoding

In the second step of data pre-processing, data encoding was performed to handle the categorical variables in the data set. Some data variables, such as 'employer size' and 'month of the injury' are categorical. To properly develop the study's ML models, all the categorical data were subjected to a widely used categorical data transformation, known as one-hot encoding (Koc et al., 2021; Li et al., 2021) to obtain binary data. An example of one-hot encoding of the 'employer size' variable is illustrated in Fig. 3. Due to one-hot encoding, the input space increased from 11 features to 59 features, thus increasing the data set's dimensionality.

3.2.3. Dimension reduction

In the third and final step of data pre-processing, the significantly increased data set dimensionality had to be reduced to minimize noise and redundancy (correlations between data points) and to speed up computation times (Ruggieri et al., 2021). Principal component analysis (PCA) is an unsupervised learning method widely used in the relevant literature to address dimensionality problems (Koc et al., 2021; Marzouk and Gaid, 2018). The PCA method breaks down large variable data sets into smaller classes of data sets without losing valuable data by applying linear transformations to the data. The result is a more precise data set reflecting the important contributing features (Zhu et al., 2021). Although using PCA can decrease classification accuracy due to some spatial information loss, it can also reduce the chance of overfitting the developed models by eliminating features with high correlations. This can provide a more reliable overall accuracy of the developed ML model (Lu et al., 2021). In this study, the PCA method was used to explain 95 % variance of the data set. By extracting the explained variance per component using the Kaiser method and eliminating principal components of eigenvalues less than 1, 39 principal components were selected.

3.3. Machine learning (ML) methods

After data pre-processing, preliminary experimentation was conducted using classification models including discriminant analysis, decision trees, logistic regression, nearest neighbors, support vectors machines and naïve bayes classifiers. Eight ML models were deemed suitable to predict injury types due to construction accidents based on the performance. These models are fine tree (FT), ensemble of boosted tree (EBT), xgboost (XGB), random forest (RF), medium gaussian support vector machine (MG SVM), quadratic support vector machine (Q SVM), one-vs rest logistic regression (OvR LR), and multinomial logistic regression (M LR). The pre-processed data was then partitioned (75 % allocated for training and 25 % for testing. The 25 % portion of the data held for validation was used to assess the model's performance in the testing phase, which assists in choosing the best model and helps prevent model overfitting (Shehadeh et al., 2021). Also, Bayesian hyperparameter optimization was used and preferred over other hyperparameter tunning (*meta*-heuristic algorithms (Fong et al., 2018), grid search (Paul et al., 2019), and random search (Bergstra and Bengio, 2012) mainly because it builds a probability model of the objective function to propose smarter choices for the next set of hyperparameters to evaluate (less validation set errors indicating better test set performance, and less number of trials indicating less time consumption) (Shahriari et al., 2015). The following subsections briefly describe the classification prediction techniques and their unique specifications in this study.

3.3.1. Decision trees

The decision tree (DT) is an analytical ML method widely used to explain complex problems in the simple hierarchical form of a top-down algorithm (Delen et al., 2017). The decision tree algorithm splits the data into two or more sets based on the most important or independent variables. The decision tree is a supervised ML method that develops a predictive model based on continuous dependent variables (Tixier et al., 2016). It is easy to understand, strong in interpretability, and suitable for processing samples with missing attributes. Decision trees can handle unrelated features and relatively quickly produce feasible and good results for large data sources (Poh et al., 2018). The decision tree's excellent accuracy and training efficiency make it an effective classification and prediction method to avoid correlations between variables (Liu and Li, 2020). It is also computationally simple, with a tree model classifier creating a large decision tree to classify the data. Based on the data values, a true/false decision must be made at each node so conclusions can be successfully achieved.

In this study, through an iterative process of experimenting and training various types of decision tree algorithms on the obtained performance results, fine tree (FT), ensemble of boosted tree (EBT), xgboost (XGB) and random forest (RF) were chosen as the suitable decision tree prediction methods (Barker et al., 2021). Hyperparameter tunning of the fine tree prediction model indicated that the maximum number of splits chosen was 100, following Gini's diversity index split criterion (Zebin et al., 2017). For the EBT model, where the preset was the boosted tree and the ensemble method was AdaBoost, the optimal number of splits was set at 20 with a learning rate of 0.1 to reduce model complexity and computational time (Liu et al., 2020). For the XGB model, the best hyperparameters indicated that sub-samples is set to 1, minimum child weight set to 5, maximum depth set to 3, and gamma set to 5. For the RF model, best hyperparameters were the minimum samples split set to 3, minimum samples leaf set to 1, and number of estimators set to 200 (Pedregosa, 2011).

3.3.2. Support vector machines (SVMs)

The support vector machine (SVM) is a widely used classification

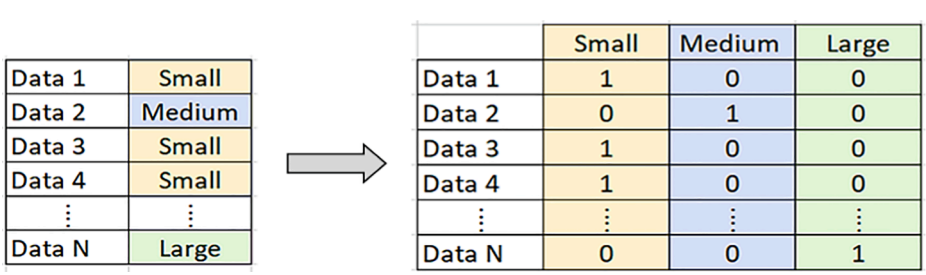

Fig. 3. Example of one-hot encoding technique performed in this study.

technique due to its excellent predictive performance and solid theoretical basis (Chen et al., 2016). The SVM counts each piece of data with points in the '*N*-dimensional' space, where N is the total number of features. Thus, each feature is a coordinate value in the *N*-dimension (Alwasel et al., 2017). The SVM then finds a straight line that distinguishes between the two groups of different data. Without relying on the entire data set, the SVM can solve high-dimensional problems and non-linear features, making it a great technique for handling types of prediction problems, such as classification, with this reflected in numerous real-world prediction problems, such as worksite accidents in the construction industry (Xu and Saleh, 2021). In this study, where multivariate analysis is the critical component of the prediction problem, using the SVM provides the desired degree of accuracy (Zhu et al., 2021). The SVM learning methods were tested on the pre-processed data, and the medium Gaussian SVM and quadratic SVM were judged as suitable prediction models for adoption. Hyperparameter tunning indicated a cubic kernel function with an optimal kernel scale of 3.3, and with a one-vs-all multi-class method (Liu et al., 2020).

3.3.3. Logistic regression (LG)

Proposed by Cox (Cox, 1958) and applied in a variety of fields, Logistic regression is widely preferred classification model specially in assessing safety risk factors. This model is useful, because it demonstrates the correlation between items and offers the advantage of interpreting and utilizing the resulting formula expressed in a simple linear form (Maalouf, 2011). It is primarily used for predicting binary dependent variables, but its extended version can also handle multi-class classification problems (Abramovich et al., 2021). In this study, One-vs-rest logistic regression (OvR LR) and multinomial logit (M LR) have been employed for the classification modeling. The best hyperparameters used for OvR LR were newton-cg solver, C (float) set to 10, and penalty l2. As for M LR, libilinear was selected as the solver, with C (float) of 0.1, and penalty l1.

4. Results

4.1. Optimum ML model to predict injury types

Performance evaluation is a primary step after running the ML algorithms, as it directly provides relevance to the different models' outcomes. Without conducting a performance evaluation on a model's outcome, results are vague, and interpretations are meaningless (Koc et al., 2021).

Various evaluation metrics are used in machine learning (ML). In this study, recall, precision, accuracy, F1 score, area under precision-recall curve (AUPRC), and the area under the receiver operating curve (AUROC) metrics are used, with these commonly used in the literature and in accident analysis applications (Kang and Ryu, 2019; Koc et al., 2021; Marzouk and Gaid, 2018; Delen et al., 2017; Lin et al., 2021; Shen et al., 2021). This section elaborates further on each metric and on its significance in the study's findings. Performance measures depend on four possible classification outcomes in multi-class classification (refer to Fig. 4): A true positive (TP), a true negative (TN), a false positive (FP), and a false negative (FN).

The study constructed a confusion matrix to assess the five models' classification performance. This matrix illustrates possible outcomes between the predicted and true conditions of the models' output (Mataei et al., 2018). As shown in Fig. 4, which represents a multi-class classifier, each row of the matrix signifies a case in an actual class, while each column embodies the cases in a predicted class. The confusion matrix details the number of TPs, TNs, FPs, and FNs for any specific class. These values are essential to determining the models' performance, such as its precision, recall, F1, accuracy, the area under the receiver operating curve (AUROC), and the area under precision-recall curve (AUPRC).

If value A is taken as an example, then the confusion matrix and the different TPs, TNs, FPs, and FNs are presented as shown in Fig. 4 below:

Predicted Values

True Values	A	B	C	D
A	Cell 1	Cell 2	Cell 3	Cell 4
B	Cell 5	Cell 6	Cell 7	Cell 8
C	Cell 9	Cell 10	Cell 11	Cell 12
D	Cell 13	Cell 14	Cell 15	Cell 16

Fig. 4. Confusion matrix for 4x4 class modelling.

$\mathbf{TP_{(A)}}$ = the value in cell 1 (green-shaded cell).

$\mathbf{FP_{(A)}}$ = the sum of values in cell 2, cell 3, and cell 4 (orange-shaded cells).

$\mathbf{FN_{(A)}}$ = the sum of values in cell 5, cell 9, and cell 13 (blue-shaded cells).

$\mathbf{TN_{(A)}}$ = the sum of values in cell 6, cell 7, cell 8, cell 10, cell 11, cell 12, cell 14, cell 15, and cell 16 (pink-shaded cells).

The rest of the models' performance metrics are dependent on the confusion matrix values and can be identified as follows:

Precision: (or the positive predictive value [PPV]) is an evaluation metric representing the ratio of correctly predicted observations under a particular class to total observations under a similar class. Simply stated, this means the proportions of positive observations that are true positives (Kang and Ryu, 2019; Delen et al., 2017):

$$\text{Precision}: PPV = \frac{TP}{TP + FP}$$

Recall: the sensitivity or true positive rate (*TPR*) is the ratio of correctly predicted observations under a particular class to all actual observations in that class. This evaluation metric measures the proportion of actual positives that are correctly identified as positives (Cheng et al., 2012):

$$\text{Recall} = TPR = \frac{TP}{TP + FN}$$

Accuracy: is defined as the ratio of the correctly classified observations to total observations. It evaluates the ratio of the correct number of predictions to the total number of samples (Wang et al., 2021):

$$\text{Accuracy} = \frac{TP + TN}{TP + FN + FP + TN}$$

F1 score: is a metric that provides an overall assessment scheme by combining recall and precision. It can be expressed as shown below and indicates the relative importance of recall versus precision. The F1 score is used when an imbalance exists in the class distribution (Wang et al., 2021); thus, the F1 score is a better metric for calculating the existing model:

$$\text{F1 score} = \frac{2 \times Precision \times Recall}{Precision + Recall}$$

AUROC: measures the area under the entire receiver operating curve (ROC). The ROC curve is plotted with the TPR (true positive rate) on the vertical axis and the false positive rate (FPR) on the horizontal axis, with

this being a robust graphical technique to determine the classification performance (Koc et al., 2021; Choi et al., 2020).. The AUROC is a performance metric widely adopted for numerous ML algorithms due to its reliability, ease of handling and interpretation (Das and Khanna, 2021; Kang, 2021; Chu et al., 2021).

AUPRC: Plots the tradeoff between precision TP/(TP + FP) (in the vertical axis) and recall TPR (in the horizontal axis) for different thresholds (Flach and Kull, 2015). A high AUPRC represents both high recall and high precision, where high precision relates to a low false positive rate, and high recall relates to a low false negative rate (Gurina, 2020). Given the imbalanced nature of the data in this study, AUPRC is the highly desirable performance metric for ML model selection. In addition, the classification threshold is set away from 0.5 to surpass the imbalanced nature of the data.

Generating the prediction models, the TP, TN, FP and FN values are collected for the eight ML methods with the values represented in the confusion matrix, as illustrated in Figs. 5 and 6. These figures integrate all values obtained from the five models in one visualization. Fig. 5 visualized an integrated confusion matrix of injury type prediction model for FN, EBT, MG SVM, and Q SVM. Fig. 6 illustrates the confusion matrix of injury type prediction model for RF, XGB, OvR LR, and M LR. The confusion matrix is generated after cross-validation. The diagonal cells show where the true class and predicted class match and that they are classified correctly.

The measures of the top four models' performance are presented in Table 3. As noted, all eight models yielded very proximate results in terms of precision, recall, F1, AUROC, accuracy, and AUPRC. However, for an imbalanced classification problem, the key metric to look at is the AUPRC. AUROC might not reflect the true performance of the model given the imbalanced nature of the data and can be misleading (Zou et al., 2016). Also, Accuracy is not a good measure for unbalanced classification problems because it gives equal importance to both false positives and false negatives which can be misleading (Chawla, 2009). Precision and recall are useful metrics subject to small positive classes being the main goal (Tahir et al., 2019). Adding to that, F1 gives equal weights to precision and recall while in practice one of them will be more important (Zou et al., 2016).. As a result, the model selection will mostly rely on AUPRC. Thus, it is noted that RF model yielded better results compared to the rest of the models. The RF yielded an average of 77.1 % precision, an average of 79.3 % accuracy for predictions across different output classes, recall average of 78.0 %, F1 score of 78.5 %, an AUROC score of 0.98, and with the AUPRC of 0.7 higher than the rest of the classifiers making it a superior model among the rest.

4.2. Feature importance

Machine learning (ML) techniques capture the complex relationships between different variables and produces accurate future prediction models. However, prediction models are incomplete without thorough model transparency through evaluating all the features that contribute the most to the model's output (Bukhsh et al., 2020). The current study evaluates features determining injury types due to construction worksite accidents. This gives more room for decision-makers to perform adequate prediction models even if some predictors are absent due to difficulties or expenses involved in data collection (Naser, 2021). Parametric sensitivity analysis examines the cause-and-effect relationship between the input and output variables, using the feature permutation importance technique on RF as an example. This method is commonly used where the mean-decrease-accuracy of the model is calculated by removing a feature at a time (Paul et al., 2013; Altmann et al., 2010; Ron-Ferguson et al., 2021). The model is trained and tested for the absence of each input variable, with the change in the model's accuracy then measured to identify the significance of the absent feature in the prediction model. The derived feature importance values are sorted and ranked in a descending order. The features' importance, derived by the permutation method, is extracted for 39 principal components, normalized as illustrated in Fig. 7.

5. Discussion

5.1. Optimum ML model to predict injury types

As shown in Fig. 5, INJ 1 (upper limbs) yielded the highest true positive (TP) classifications using EBT; followed by 1368 TP classifications for INJ 2 (lower limbs); 1762 TP classifications for INJ 3 (head and neck); and 1675 TP classifications for INJ 4 (back and truck) all with RF.

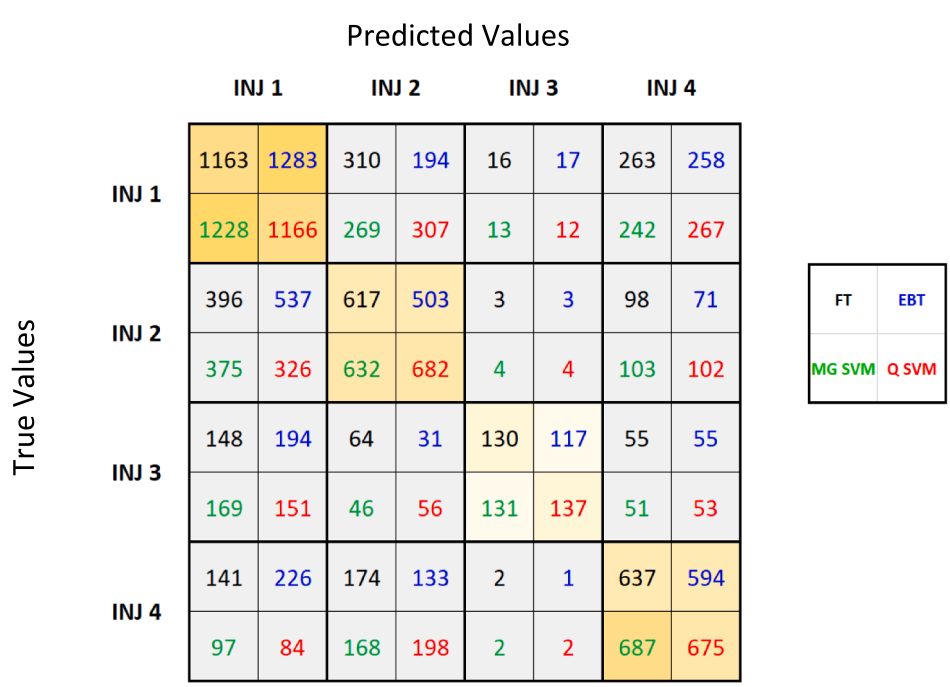

Fig. 5. Integrated confusion matrix of injury type prediction model (INJ 1 = upper limbs; INJ 2 = lower limbs; INJ 3 = head and neck; INJ 4 = back and trunk; FT = fine tree; EBT = ensemble of boosted tree; MG SVM = medium Gaussian support vector machine; Q SVM = quadratic support vector machine).

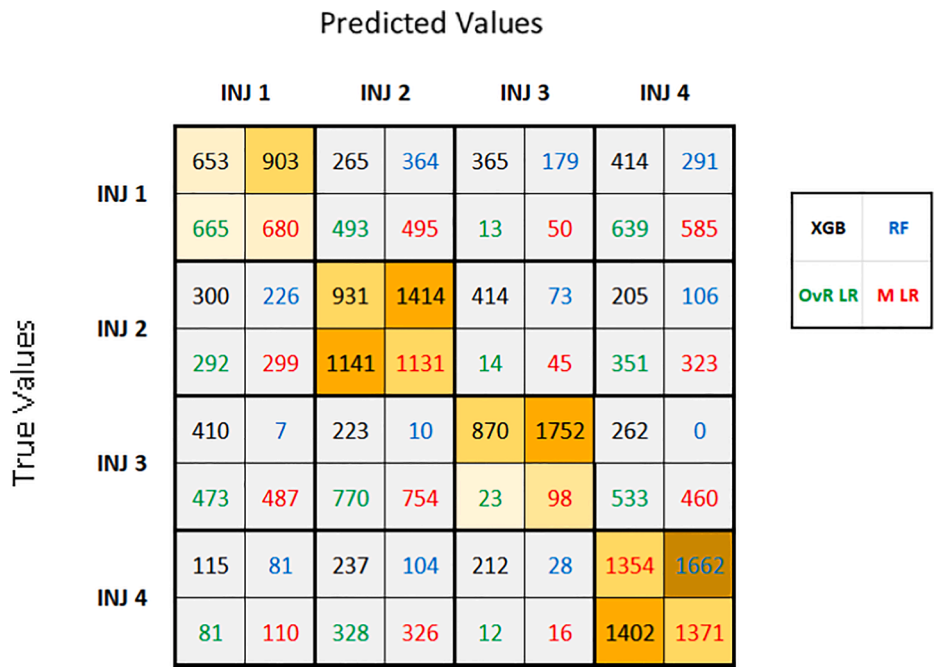

Fig. 6. Integrated confusion matrix of injury type prediction model (INJ 1 = upper limbs; INJ 2 = lower limbs; INJ 3 = head and neck; INJ 4 = back and trunk; RF = Random Forest, XGB = xgboost; OvR LR = one-vs -rest logistic regression; MLR = multinomial logistics regression)).

Table 3
Performance metrics results for the eight ML models.

	Decision tree				Support vector machine		Logistic Regression	
	RF	**FT**	**EBT**	**XGB**	**MG SVM**	**Q SVM**	**OvR LR**	**M LR**
AUPRC	0.78	0.66	0.48	0.55	0.46	0.43	0.45	0.45
Precision	77.1	75.5	65.3	51.8	68.3	68.1	44.0	59.9
Recall	78.0	76.2	52.6	52.1	58.0	58.3	59.4	60.3
F1 score	78.5	75.2	55.1	51.8	60.0	60.1	50.6	53.4
Accuracy	79.2	76.5	59.2	52.6	63.5	63.1	44.6	45.6
AUROC	0.98	0.96	0.77	0.77	0.80	0.79	0.62	0.70

However, these values are irrelevant without a meaningful evaluation. Thus, six performance metrics (precision, recall, F1 score, accuracy, AUPRC, and AUROC) were used in this study to assess the prediction technique used. The RF yielded the best results with an average of 77.1 % precision, an average of 79.3 %accuracy for predictions across different output classes, recall average of 78.0 %, F1 score of 78.5 %, AUROC score of 0.98, and most importantly AUPRC of 0.78. These findings address the first research question, which sought to determine the optimum ML model for predicting injury types due to construction worksite accidents.

The predicted classes of worksite accident consequences on injury types of 'lower limbs', 'upper limbs', 'head and neck', and 'back and trunk' provide vital information for assessing the risk level of different activities at construction workplaces and, accordingly, for undertaking proactive accident prevention measures at construction workplaces and providing adequate personal protective equipment (PPE) (Sarkar et al., 2020). In addition, the findings assist insurance providers with accurate predictions. The findings can also be mapped against lost workdays related to the different injury types, and appropriate safety controls can be developed accordingly to reduce schedule overruns in construction projects.

5.2. Feature importance

The feature importance technique was used to assign a score to each attribute based on its significance to the prediction power. As shown in Fig. 7, different features determined the injury types resulting from construction worksite accidents. For features determining the 'upper limbs' injury type, the top determinants are related to muscle/joint/ligament injuries; accidents at the beginning of the week (Mondays and Tuesdays); age groups of 40–49 and 60+; workers in environmental and animal agencies, and in heavy civil and installation services sub-sectors; and small employers. According to Antwi-Afari, et al. (Antwi-Afari et al., 2017), muscle and joint fatigue injuries were significant factors in upper limb injuries. It should be noted that individual level of analysis features other than age, such as gender, hours worked per week and employment type, were not the main determinants of upper limb injuries.

Features determining the 'lower limbs' injury type were related to muscle/joint/ligament and musculoskeletal accidents; age group 30–59; body stressing accident mechanism; accidents occurring on Tuesdays; residential building sub-sectors; and unpowered tools, materials and substances agencies. Some of these findings were similar to those of Ekpenyong and Inyang (Ekpenyong and Inyang, 2014), in which lower limb injuries were associated with indirect symptoms of the back and muscles' musculoskeletal components connected to lower limbs (Ekpenyong and Inyang, 2014). As with upper limbs injury types, features at the individual level of analysis, other than age, such as gender, hours worked per week and employment type, were not the main determinants of lower limbs injury types.

For features determining the 'head and neck' injury type due to construction worksite accidents, the most significant features were falls, sound and pressure accidents, accident natures of fractures,

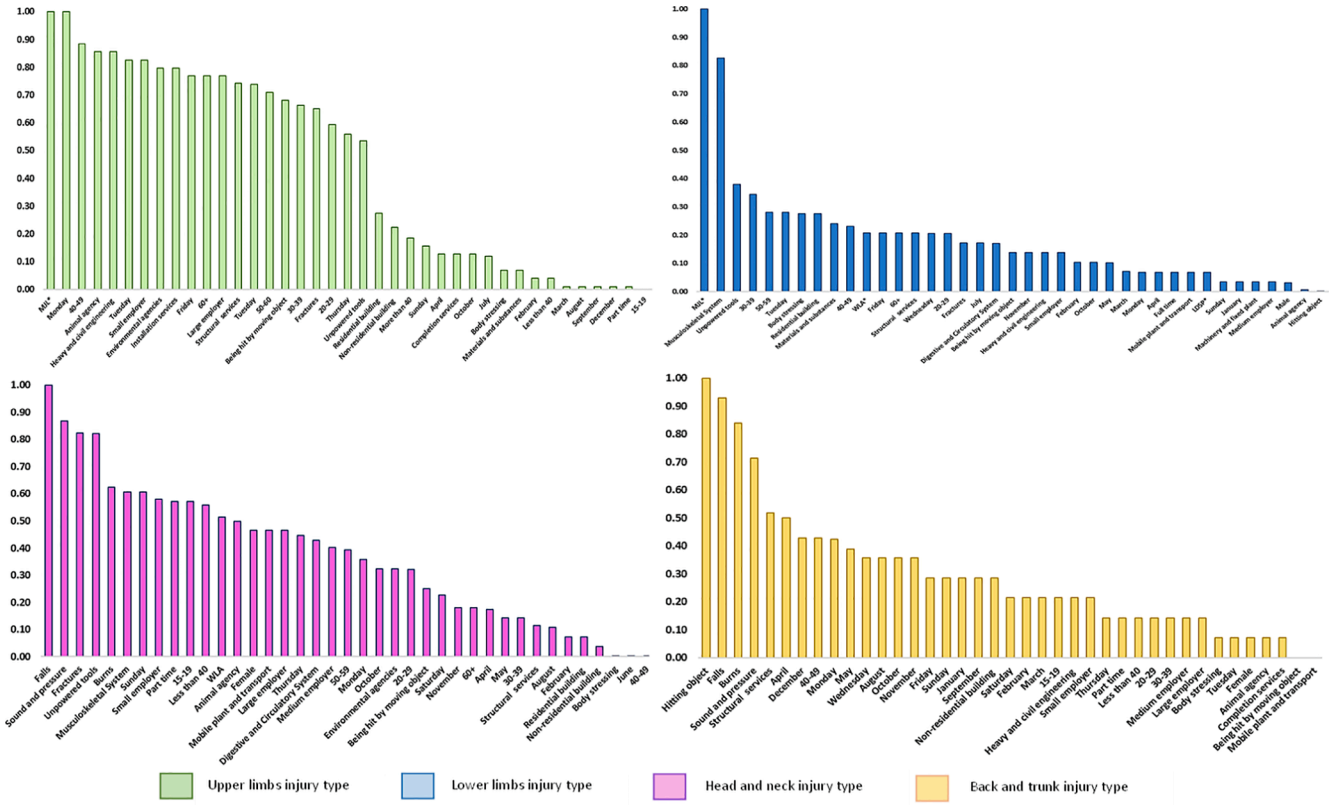

Fig. 7. Feature importance of different IT using RF model.

musculoskeletal and burns, unpowered tools agencies, small employer size, and works falling on Sundays. These findings relate to increased correlations between falls in construction and head and neck injuries (Brolin et al., 2021; Anwer et al., 2021). Again, features at the individual level of analysis, other than age, such as gender and employment type, were not the main features of head and neck injuries.

Finally, features determining 'back and trunk' injury types due to construction worksite accidents were accident mechanisms of falls, hitting objects and body stressing; musculoskeletal accident nature; structural services sub-sector; age group 40–49, accidents at the beginning of the week (Monday); and accidents in April, May, and December.

Accident mechanisms and nature were the most significant factors determining injury types in the construction and infrastructure industries. The results are comparable to those in a study undertaken by Kang and Ryu (Kang and Ryu, 2019) which indicated that features of some individual levels of analysis (e.g. gender, employment type, hours worked) are not significant in determining accident mechanisms (e.g., falls, being hit by moving objects, etc.). Nevertheless, some factors like injuries occurring on the beginning of the week 'Mondays', and 'age groups between (30–49)' were significant factors contributing to the different injury types regardless of their category. On the other hand, some features were specific to certain injury types, such as 'sounds and pressure' feature and 'head and neck' related injury type which can be directly relatable as high sounds working environments can cause head related injurie and disorders such as noise induced hearing loss (Mirza et al., 2018) (Nadhim et al., 2016). Also, 'falls' feature and 'back and trunk' injury type and 'head and neck' injury types can also be directly associated as presented in (Terayama et al., 2022) and (Singh et al., 2020) with high spinal cord injuries and head fractures. The findings reported in this section answer the second research question on what features determine the accuracy of the different injury type prediction models.

The feature importance technique provides decision-makers hands-on information on important features to include in their predictive models. It also encourages insurance policy makers to evaluate their insurance and compensation schemes based on the different attributes that determine the injury type. There are practical implications for construction sites when injury types are predictable (Winge and Albrechtsen, 2018). Increased safety supervision is critically important for works at the start of the week (Sundays and Mondays) due to potential disorientation and distractions. As another practical implication, targeted safety training can benefit certain age groups (30–59), and certain industries, including residential and heavy civil sub-sectors. Also, musculoskeletal injury is found to be the top 5 factor in 3 out of 4 injury types, thus adequate PPE should be provided for activities involving working from heights as an example. Also, proper site's hazards screening, and adequate safety trainings should be strongly enforced. The findings of this research are in line with previous studies highlighting the importance of implementing more than one safety control to avoid different injury types. For instance, (Subedi and Pradhananga, 2021) created a moment prediction model with Musculoskeletal injuries data using random forest algorithm to monitor safe work posture, and safety frontier in high risk construction activities. The computed safety frontier potentially aids safety managers to enhance safety strategies by providing a higher safety benchmark in monitoring construction sites. Moreover, (Breloff et al., 2019) highlighted the impact of high risk construction work settings on Musculoskeletal injuries. The findings facilitate effective interventions such as education, training, and tools to minimize repetitive works in construction tasks.

6. Conclusion

This study investigated different ML algorithms to optimally predict injury types due to worksite accidents. The data set of construction accidents in Victoria, Australia, from 2016 to 2021 was pre-processed to provide appropriate input to different ML algorithms. The outputs of predictive models were evaluated against six performance metrics: accuracy, prediction, F1 score, AUROC, AUPRC, and recall. Then, features

determining the accuracy of the ML models were analyzed using the permutation feature importance method. The findings of the prediction models indicated the superiority of RF in predicting injury types due to construction worksite accidents. Moreover, features related to accident nature and mechanism were found to be critical in most predictive models. This study contributes to the safety literature and practice by deploying ML methods to predict injury types, which can enhance construction safety.

Some limitations in this study need to be highlighted. Firstly, the main focus is on the construction and infrastructure industries and other industries such as manufacturing were left out of the scope. Also, construction-site related features such as projects length and cost, and individual-specific features like how many years the individual has been working in the construction industry, and what type of PPEs have been provided to them can potentially add value to safety research, which serves as a potential direction to future research and researchers. Moreover, while this study used ML models to predict injury types, integration with other methods, such as time series analysis can enhance prediction performance in real-life project settings. A potential direction for future research is deploying deep learning methods to process big data and predict construction injuries. Future research could also focus on predicting accident severity and consequences in an attempt to enhance construction safety.

CRediT authorship contribution statement

Maryam Alkaissy: Writing – original draft, Methodology, Formal analysis, Conceptualization. **Mehrdad Arashpour:** Conceptualization, Funding acquisition, Data curation, Methodology, Supervision. **Emadaldin Mohammadi Golafshani:** Writing – original draft, Visualization, Software. **M. Reza Hosseini:** Writing – review & editing, Project administration, Software. **Sadegh Khanmohammadi:** Validation, Writing – review & editing. **Yu Bai:** Writing – review & editing, Validation. **Haibo Feng:** Writing – review & editing, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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